Temporal Prototype Alignment

for Frozen-Feature Dataset Distillation

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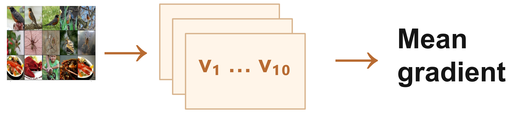
NeurIPS 2026

Project page

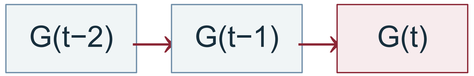
1 Why temporal reuse?

LGM distills images through frozen encoders. It averages ten augmented views per step to stabilize classifier gradients. The concurrent views increase memory and I/O costs.

LGM: evidence across space



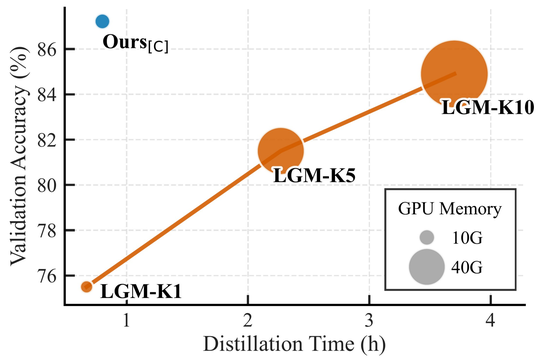
TPA: evidence across time



One fresh view + a running prototype

Fixed-size moments reuse gradient history across steps instead of concurrent views.

Memory and accuracy

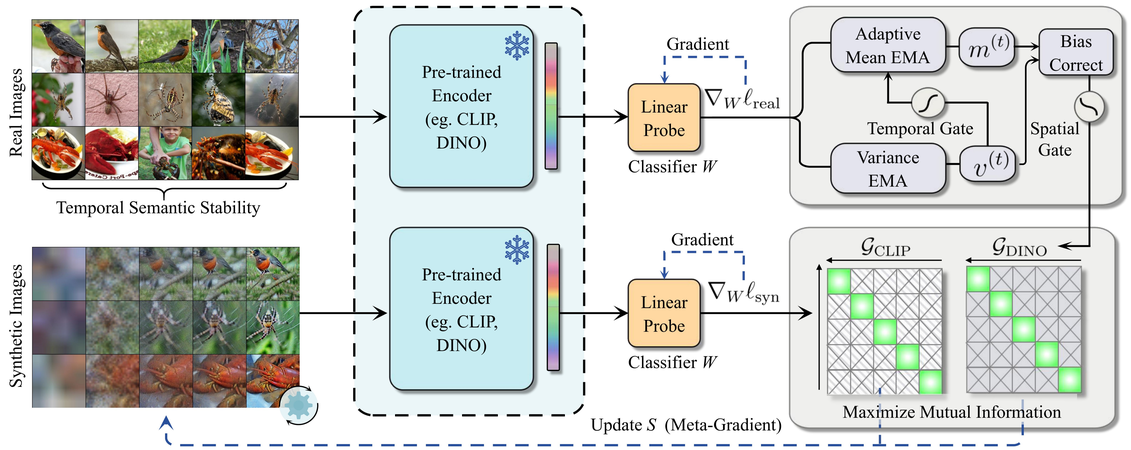


81 → 7 GB **3.6 → 0.8 h** **84.9 → 86.7%**

CLIP, ImageNet-100, IPC = 1. LGM: 4 × RTX 4090; TPA: 1 × RTX 4090. Reported hardware differs.

2 How temporal prototypes guide synthesis

Real gradients build temporal prototypes. Synthetic gradients align with these targets through frozen teachers; only synthetic pixels are updated.



Upper path: temporal prototypes. Lower path: teacher-local class alignment.

Adaptive Temporal Momentum (ATM)

ATM tracks first and second moments of single-view gradients. A **reliability-dependent gate** balances new observations with history; **inverse-RMS weighting** forms the temporal prototype. Initialization-mass correction accounts for zero-initialized moments. Real and synthetic batches share the same freshly sampled head at each step.

Contrastive Consensus Alignment (CCA)

CCA compares class-wise synthetic gradients with real prototypes **within each teacher**. Matched classes are positives; other classes are negatives. Summing the CLIP, DINOv2 and MoCo-v3 losses optimizes **one shared image set**. Teachers retain separate heads and prototype banks, so feature dimensions may differ and no dot product crosses teacher spaces.

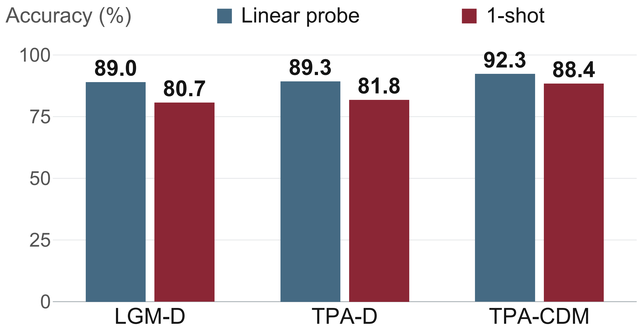
3 Cross-encoder transfer

ImageNet-100, one image per class (IPC = 1). We evaluate one jointly distilled set on nine unseen encoders using linear probing and training-free 1-shot inference.

+3.3 pp
Linear probing

+7.7 pp
1-shot inference

Gain over LGM-D · mean of nine unseen encoders



TPA-CDM reaches **92.3%** linear-probe and **88.4%** 1-shot accuracy. Gains over LGM-D extend beyond the teachers used for synthesis and cover both evaluation protocols.

What does the distilled set look like?



Selected images from the paper's TPA-CDM set.